

3.5 Trig. Derivatives (requires 3.3)

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Differentiate the following functions and fully simplify.

a) $g(t) = \frac{t^3}{\sec(t)} + \csc(t)$

$$g(t) = t^3 \cos(t) + \csc(t)$$

$$g'(t) = 3t^2 \cos(t) + t^3(-\sin(t)) - \csc(t) \cot(t)$$

$$g'(t) = 3t^2 \cos(t) - t^3 \sin(t) - \csc(t) \cot(t)$$

b) $y = \sec(x) \cot(x) \sin(x) \cos(x) \csc(x)$

$$y = \frac{1}{\cos(x)} \cdot \frac{\cos(x)}{\sin(x)} \cdot \frac{\sin(x)}{1} \cdot \frac{\cos(x)}{1} \cdot \frac{1}{\sin(x)} = \frac{\cos(x)}{\sin(x)} = \cot(x)$$

$$y' = -\csc^2(x)$$

c) $h(x) = e^x \sin(x) \tan(x)$

$$h'(x) = e^x \sin(x) \tan(x) + e^x \cos(x) \tan(x) + e^x \sin(x) \sec^2(x)$$

$$h'(x) = e^x \sin(x) \tan(x) + e^x \sin(x) + e^x \sin(x) \sec^2(x)$$

$$h'(x) = e^x \sin(x) [\tan(x) + 1 + \sec^2(x)]$$

d) $f(x) = \frac{\sin(x)}{1 + \cos(x)}$

$$f'(x) = \frac{[1 + \cos(x)] \cos(x) - [-\sin(x)] [\sin(x)]}{[1 + \cos(x)]^2} = \frac{\cos(x) + \cos^2(x) + \sin^2(x)}{[1 + \cos(x)]^2} = \frac{\cos(x) + 1}{[1 + \cos(x)]^2} = \frac{1}{1 + \cos(x)}$$

2. Use your knowledge that $\frac{d}{dx}[\sin(x)] = \cos(x)$ to show that $\frac{d}{dx}[\csc(x)] = -\csc(x) \cot(x)$.

$$\frac{d}{dx}[\csc(x)] = \frac{d}{dx}\left[\frac{1}{\sin(x)}\right] = \frac{\sin(x) \cdot 0 - \cos(x)}{\sin^2(x)} = \frac{-\cos(x)}{\sin^2(x)} = -\frac{1}{\sin(x)} \cdot \frac{\cos(x)}{\sin(x)} = -\csc(x) \cot(x)$$

3. If $H(x) = x \cdot \sec(x)$, find $H'(x)$ and $H''(x)$.

$$H'(x) = \sec(x) + x \sec(x) \tan(x)$$

$$H''(x) = \sec(x) \tan(x) + \sec(x) \tan(x) + x \sec(x) \tan^2(x) + x \sec^3(x)$$

4. Find the second derivative $f''(t)$ of the function $f(t) = \csc(t)$.

$$f'(t) = -\csc(t) \cot(t)$$

$$\begin{aligned} f''(t) &= -[-\csc(t) \cot(t)] \cot(t) + -\csc(t) [-\csc^2(t)] \\ &= \csc(t) \cot^2(t) + \csc^3(t) \\ &= \csc(t) [\cot^2(t) + \csc^2(t)] \end{aligned}$$

5. Suppose $f\left(\frac{\pi}{3}\right) = 4$ and $f'\left(\frac{\pi}{3}\right) = -2$, and let $g(x) = f(x) \cdot \sin(x)$ and $h(x) = \frac{\cos(x)}{f(x)}$.

- a) Find $g'\left(\frac{\pi}{3}\right)$

$$g'(x) = f'(x) \sin(x) + f(x) \cos(x)$$

$$g'\left(\frac{\pi}{3}\right) = f'\left(\frac{\pi}{3}\right) \sin\left(\frac{\pi}{3}\right) + f\left(\frac{\pi}{3}\right) \cos\left(\frac{\pi}{3}\right) = (-2) \left(\frac{\sqrt{3}}{2}\right) + (4) \left(\frac{1}{2}\right) = -\sqrt{3} + 2$$

- b) Find $h'\left(\frac{\pi}{3}\right)$

$$h'(x) = \frac{-f(x) \sin(x) - f'(x) \cos(x)}{[f(x)]^2}$$

$$h'\left(\frac{\pi}{3}\right) = \frac{-f\left(\frac{\pi}{3}\right) \sin\left(\frac{\pi}{3}\right) - f'\left(\frac{\pi}{3}\right) \cos\left(\frac{\pi}{3}\right)}{\left[f\left(\frac{\pi}{3}\right)\right]^2} = \frac{-(4) \left(\frac{\sqrt{3}}{2}\right) - (-2) \left(\frac{1}{2}\right)}{(4)^2} = \frac{-2\sqrt{3} + 1}{16}$$

6. What is the equation of line tangent to the graph of $y = 2 \tan(x) + 3x^2 + x + 7$ at its y -intercept?

$$y(0) = 2 \tan(0) + 3(0)^2 + (0) + 7 = 7$$

So the y -intercept occurs at the point $(0, 7)$.

$$\text{We then calculate: } y' = 2 \sec^2(x) + 6x + 1$$

$$\text{The slope of the tangent line is then given by: } y'(0) = 2 \sec^2(0) + 6(0) + 1 = 2 + 1 = 3$$

So the equation of the tangent line is given by:

$$y - 7 = 3(x - 0)$$

$$y = 3x + 7$$

7. For what values of x does the graph of $f(x) = x + 2 \sin(x)$ have a horizontal tangent?

$$\text{We calculate: } f'(x) = 1 + 2 \cos(x)$$

$$\text{We need to know when } 1 + 2 \cos(x) = 0 \rightarrow \cos(x) = -\frac{1}{2}$$

$$\text{This occurs when } x = \frac{2\pi}{3} + 2\pi k \text{ or } x = \frac{4\pi}{3} + 2\pi k, \text{ for some integer } k.$$

Note: This means that there are infinitely many horizontal tangents on the graph of $f(x)$.